

Use Emscripten to Compile Code Into JavaScript

This article gives an introduction to Emscripten, an LLVM to the JS compiler.



Imagine that you have a very large project written in C/C++, and you now want it in JavaScript, so as to port it to the Web. You might ask, “Why in JavaScript?” Well, you can run applications written in high-level languages, but you need to install many plug-ins for that and it doesn’t always work as expected. JavaScript is currently supported in almost all Web browsers. With the coming of Node.js and MongoDB, and an array of other JavaScript-based front-ends and back-ends, it makes sense to port your favourite app. Rewriting the code in JavaScript is too tedious, so Emscripten is the tool that you are looking for.

Emscripten is an open source LLVM (low-level virtual machine) to the JavaScript compiler, using which you can compile C and C++ code into JavaScript! This avoids the need to rewrite code.

Getting started

Some of the requirements are the Emscripten code, LLVM with Clang, Node.js and Python 2.6. You need to clone the Emscripten source code, available at [github/kripken](https://github.com/kripken). You can use your favourite git client or the CLI, with `git clone git://github.com/kripken/emscripten.git`

LLVM with Clang (3.1 is the officially supported version): The LLVM Project is a collection of compiler and tool-chain technologies, and Clang is a new C/C++ compiler being developed on top of LLVM. So why don't we use GCC? Well, Clang can be a good replacement for GCC. Clang seems to have good compilation speed—nearly twice the speed of GCC. Also, Clang provides better comments if your code fails at some point—so it doesn't take much time to navigate through the code and fix it. Here are the steps to build and install Clang:

```
$ mkdir /path/to/clang-build
$ cd /path/to/clang-build
$ svn co http://llvm.org/svn/llvm-project/llvm/trunk llvm
$ cd llvm/tools
$ svn co http://llvm.org/svn/llvm-project/cfe/trunk clang
$ cd ../../
$ mkdir build
$ cd build
$ ../llvm/configure -enable-optimized -disable-assertions
$ make && sudo make install
```

Node.js (0.5.5 or above): To run the back-end JavaScript, it needs to be interpreted and executed, which is what Node.js does. We can clone Node.js from the git client:

```
$ git clone https://github.com/joyent/node.git
```

After cloning, you need to build it, as follows:

```
$ cd node
$ ./configure
$ make
$ sudo make install
```

Now that you are ready with the prerequisites, set up Emscripten. Check if Clang and Node work perfectly—go to the Emscripten directory, and run the following command:

```
$ clang tests/hello_world.cpp
$ ./a.out
```